

FATIH KURT

AI Engineer

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SUMMARY

AI Engineer focused on production LLM systems: RAG pipelines, chat agents, multi-agent orchestration and process automation. Currently building internal AI tooling at Siemens, on a foundation of software and data engineering at BMW.

EXPERIENCE

AI Engineer, Siemens

Working student, AI tooling for an engineering department

02/2026 – present

Nuremberg

- Built and deployed a **RAG system** over the department's technical standards corpus – covering ingestion, chunking, retrieval from a **Chroma** vector database, and answer generation through the internal Siemens LLM API. Evaluated chunking and retrieval settings against real user queries before rollout.
- Built a **chat agent in Microsoft Teams** connected to the internal knowledge base: the agent answers incoming questions about lab services directly in the channel, taking over requests that engineers previously answered manually.
- Built an **email classification pipeline** using LLM prompts with HTML-to-text preprocessing, implemented in **n8n**. Proposed the project myself, quantified the time saved, and presented the business case within the department.
- Wrote a **meeting summariser** in Python with transcript chunking for recordings of **60+ minutes**, returning structured action items into the team workflow.
- Automated recurring department processes in **Power Automate**.

Data Engineer, BMW

Intern → working student, BMW & MINI digital customer channels

08/2024 – 02/2026

Munich

- Automated data pipelines on **AWS** and **Snowflake** drawing from **30+ source systems** into roughly **200 tables**; ran SQL and Python analyses to surface trends behind product decisions.
- Built and maintained **KPI dashboards** in Tableau, Power BI and Streamlit for the BMW and MINI digital customer channels, used by around **150 people** across several agile product teams.
- Wrote a **monitoring script for test results** with email alerting, so failures surface immediately rather than during manual checks.
- Built **Streamlit** applications for data exploration and refactored existing code for performance and maintainability.

Embedded Software Engineer, Continental

Working student, embedded systems

02/2024 – 08/2024

Ingolstadt

- Wrote and optimised **C/C++ modules** for the **MX8 board** in Head-Up Display systems.

Python Test Automation Engineer, Magna

Working student, System Validation Driving Functions & E-Mobility

06/2023 – 02/2024

Ingolstadt

- Wrote executable **Python test cases** from test designs and maintained specifications in **DOORS** for safety-relevant vehicle functions.

PROJECTS

Dual-Agent Harness: Coder and Tester Agent | LangGraph · LangChain · Docker · OpenRouter · Python

2026

- Personal learning project on agent systems: a **coder agent** implements a REST API specification inside a **Docker sandbox with no network access**, while a **tester agent** probes the running API black-box over HTTP against the same specification, without ever seeing the code.
- Round orchestration built as a **LangGraph** with SQLite checkpointing for resumable runs. Token usage, change scope and golden-test results measured per run; ledger validation and rollback on rule violations.

- **Multilingual chatbot** for hotels with **multi-tier LLM routing** to optimise inference costs. Development completed and deployed.

Bachelor Thesis: RL for Adaptive Traffic Signal Control | Python · SUMO/TraCI · PPO · MLflow

10/2025 – 03/2026

- Framework where **PPO agents** and human players control traffic signals side by side in SUMO. Tracked training runs and hyperparameter sweeps in **MLflow** and evaluated agent performance against the human baseline across scenarios. Grade **1.3**.

Autonomous Driving Team Project, THI | ROS 2 · Docker · Gazebo · C++

04/2025 – 08/2025

- Real-time **time-to-collision ROS 2 nodes** with perception integration, tested in simulation and on the actual vehicle; also built the team’s Docker setup for reproducible installs.

Bosch Mobility Challenge & Teknofest Racing | OpenCV · YOLO

2022 – 2024

- Road detection and real-time object recognition for autonomous vehicle perception.

SKILLS

LLM & Agent Engineering: RAG (chunking, embeddings, retrieval), LangChain, LangGraph, multi-agent systems, Chroma, OpenRouter, Ollama, prompt engineering, LLM evaluation

MLOps & Deployment: MLflow (experiment tracking), Langfuse (tracing), Docker, REST API deployment, Git/GitHub, automated monitoring & alerting

Automation: n8n, Microsoft Power Automate, Microsoft Teams bots, Telegram Bot API, workflow design

Programming: Python (FastAPI, REST APIs), C/C++, SQL

Data & Cloud: AWS, Snowflake, Power BI, Streamlit, Tableau

Systems & Robotics: Linux, Yocto, ROS 2, Gazebo, SUMO

ML / Computer Vision: Reinforcement Learning (PPO), YOLO, OpenCV, TensorFlow

EDUCATION

B.Eng. Autonomous Vehicle Engineering, Technische Hochschule Ingolstadt

Thesis (Grade: **1.3**): Reinforcement Learning for adaptive traffic signal control

10/2022 – 07/2026

Ingolstadt

German Language Courses, Universität des Saarlandes

10/2021 – 08/2022

Saarbrücken

LANGUAGES & CERTIFICATIONS

Languages: Turkish (native), German (C1), English (C1)

Certifications: Machine Learning Specialization (Coursera): supervised & unsupervised learning, recommender systems, RL; MATLAB Onramp (MathWorks)